

Calculation Table of Contents

Temperature Rise Report (IEC 60890)	3
IEC 60890 – Calculation Preconditions and Compliance	3
Temperature Rise - Tier Summary	4
1 Tier — Tier 1	5
Components	5
1.1 Temperature Rise Summary — Tier 1	5
1.1.1 Temperature Rise Curve	5
1.1.2 Temperature Rise Results	5
1.2 IEC 60890 Calculation Sheet — Tier 1	6
1.2.1 Effective cooling surfaces	6
1.2.2 IEC 60890 calculation parameters	6
1.2.3 Enclosure heat dissipation and ventilation	6
2 Tier — Tier 2	7
Components	7
2.1 Temperature Rise Summary — Tier 2	7
2.1.1 Temperature Rise Curve	7
2.1.2 Temperature Rise Results	7
2.2 IEC 60890 Calculation Sheet — Tier 2	8
2.2.1 Effective cooling surfaces	8
2.2.2 IEC 60890 calculation parameters	8
2.2.3 Enclosure heat dissipation and ventilation	8
2.3 Component Current Capacity Assessment - Tier 2	8
2.3.1 Component derated current capacity	8
3 Tier — Tier 3	9
Components	9

3.1 Temperature Rise Summary — Tier 3 9

 3.1.1 Temperature Rise Curve 9

 3.1.2 Temperature Rise Results 9

3.2 IEC 60890 Calculation Sheet — Tier 3 10

 3.2.1 Effective cooling surfaces 10

 3.2.2 IEC 60890 calculation parameters 10

 3.2.3 Enclosure heat dissipation and ventilation 10

3.3 Component Current Capacity Assessment - Tier 3 10

 3.3.1 Component derated current capacity 10

Temperature Rise Report (IEC 60890)

IEC 60890 – Calculation Preconditions and Compliance

Item	Assessment Condition	Compliance
------	----------------------	------------

Temperature Rise - Tier Summary

Tier	Compliance	Cooling arrangement	Airflow (m ³ /h)
Tier 1	Compliant	Natural convection	-
Tier 2	Compliant	Forced ventilation	960
Tier 3	Compliant	Forced ventilation	480

Important application of results

A compliant result means the calculated internal air temperatures are within the tier limits for the cooling arrangement shown. Natural-convection results use enclosure heat dissipation under the specified conditions. Forced-ventilation results include the selected fan and are conditional on providing and maintaining the stated delivered airflow through that tier. Enclosure temperature compliance does not establish full rated-current capacity for components. Apply the component current capacities and temperature limits reported for each tier.

- **Tier 2:** Selected fan: 3245800; delivers 960 m³/h of airflow.
- **Tier 3:** Selected fan: Custom fan arrangement; delivers 480 m³/h of airflow.

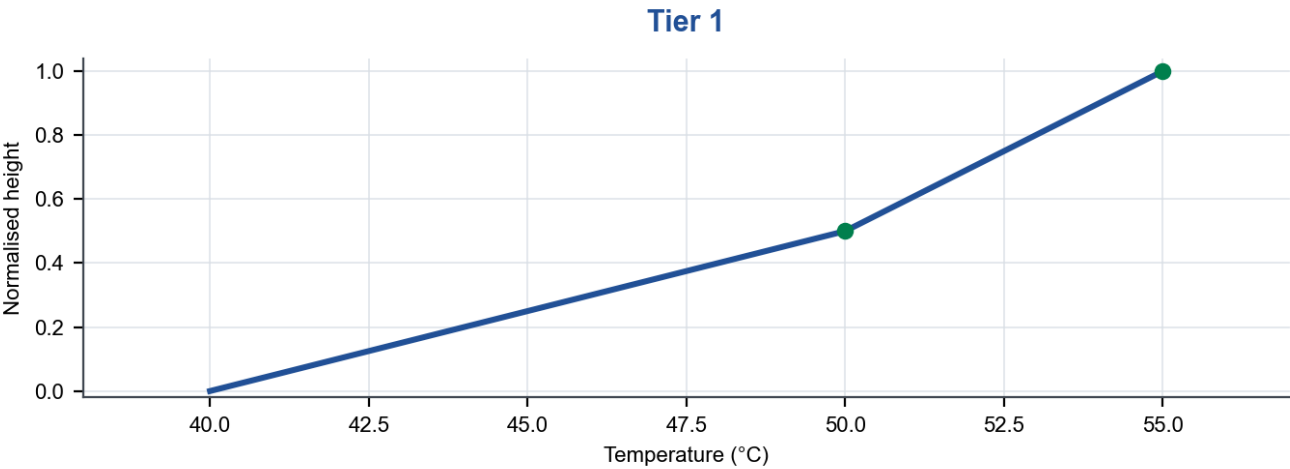
1 Tier — Tier 1

Components

Description	Part No.	Qty	W each	Total (W)
Control power supply	D100	1	100.0	100.0
Tier heat subtotal (W)			300.0	

1.1 Temperature Rise Summary — Tier 1

1.1.1 Temperature Rise Curve



1.1.2 Temperature Rise Results

Parameter	Value
Tier heat load (W)	300.0
Effective cooling area Ae (m²)	4.500
Final Temp @ 0.5t (°C)	50.0
Final Temp @ 1.0t (°C)	55.0
Maximum Allowed (°C)	70.0
Compliance @ 0.5t	Compliant
Compliance @ 1.0t	Compliant

Important application of results

The calculated temperatures use natural enclosure heat dissipation under the specified conditions. A non-compliant result requires additional cooling or a design change. Enclosure temperature compliance does not establish full rated-current capacity; component current capacities and temperature limits also apply.

1.2 IEC 60890 Calculation Sheet — Tier 1

The following tables summarise the IEC 60890 enclosure thermal calculation including effective cooling surfaces, enclosure constants, and heat dissipation balance.

1.2.1 Effective cooling surfaces

Surface	Dimensions (m)	A0 (m ²)	b	Ae (m ²)
Front	0.60 × 1.80	1.08	0.90	0.97
			Total Ae	4.50

1.2.2 IEC 60890 calculation parameters

Parameter	Value
Effective area Ae (m ²)	4.500
k (enclosure constant)	0.150
c (distribution factor)	1.400
x (exponent factor)	0.804
f (Ae > 1.25 m ²)	1.000
Ambient temperature (°C)	40.0

1.2.3 Enclosure heat dissipation and ventilation

Parameter	Value
Is vented (openings specified)	No
Vent inlet opening area (cm ²)	N/A
Maximum natural enclosure dissipation P890 (W)	180.0 W
Heat for ventilation / cooling (W)	120.0 W
Required airflow (m ³ /h)	—
Horizontal partitions	Disabled

2 Tier — Tier 2

Components

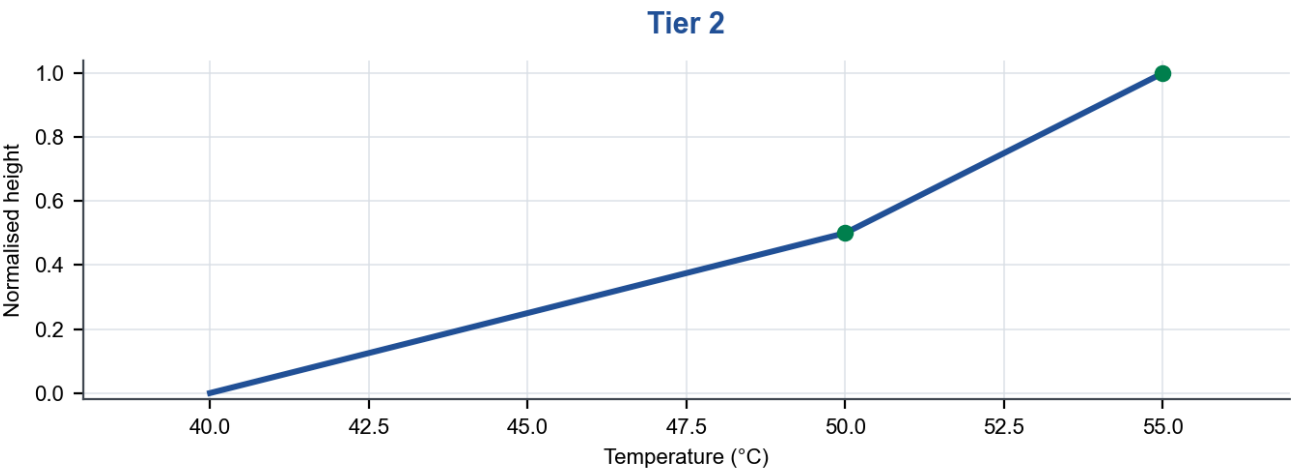
Description	Part No.	Qty	W each	Total (W)
Variable speed drive	D100	1	100.0	100.0

Tier heat subtotal (W)

300.0

2.1 Temperature Rise Summary — Tier 2

2.1.1 Temperature Rise Curve



2.1.2 Temperature Rise Results

Parameter	Value
Tier heat load (W)	300.0
Effective cooling area Ae (m²)	4.500
Final Temp @ 0.5t (°C)	50.0
Final Temp @ 1.0t (°C)	55.0
Maximum Allowed (°C)	70.0
Compliance @ 0.5t	Compliant
Compliance @ 1.0t	Compliant

Important application of results

Compliance is conditional on providing and maintaining the stated delivered airflow through the tier. A non-compliant result requires additional cooling or a design change. Enclosure temperature compliance does not establish full rated-current capacity; component current capacities and temperature limits also apply.

- **Tier 2:** Selected fan: 3245800; delivers 960 m³/h of airflow.

2.2 IEC 60890 Calculation Sheet — Tier 2

The following tables summarise the IEC 60890 enclosure thermal calculation including effective cooling surfaces, enclosure constants, and heat dissipation balance.

2.2.1 Effective cooling surfaces

Surface	Dimensions (m)	A0 (m²)	b	Ae (m²)
Front	0.60 × 1.80	1.08	0.90	0.97
			Total Ae	4.50

2.2.2 IEC 60890 calculation parameters

Parameter	Value
Effective area Ae (m²)	4.500
k (enclosure constant)	0.150
c (distribution factor)	1.400
x (exponent factor)	0.804
f (Ae > 1.25 m²)	1.000
Ambient temperature (°C)	40.0

2.2.3 Enclosure heat dissipation and ventilation

Parameter	Value
Is vented (openings specified)	No
Vent inlet opening area (cm²)	N/A
Maximum natural enclosure dissipation P890 (W)	180.0 W
Heat for ventilation / cooling (W)	120.0 W
Required airflow (m³/h)	—
Horizontal partitions	Disabled
Selected fan	3245800
Delivered airflow (m³/h)	960

2.3 Component Current Capacity Assessment - Tier 2

2.3.1 Component derated current capacity

Component	Rated (A)	Internal temp (°C)	Derated output (A)
Variable speed drive*	100.0	55.0	85.0

* Component(s) temperature deratings were solved directly using manufacturer data. Selected fan: 3245800; delivers 960 m³/h of airflow.

Current capacity evaluated at solved top air temperature (55.0°C). Component temperature ratings must also be satisfied.

3 Tier — Tier 3

Components

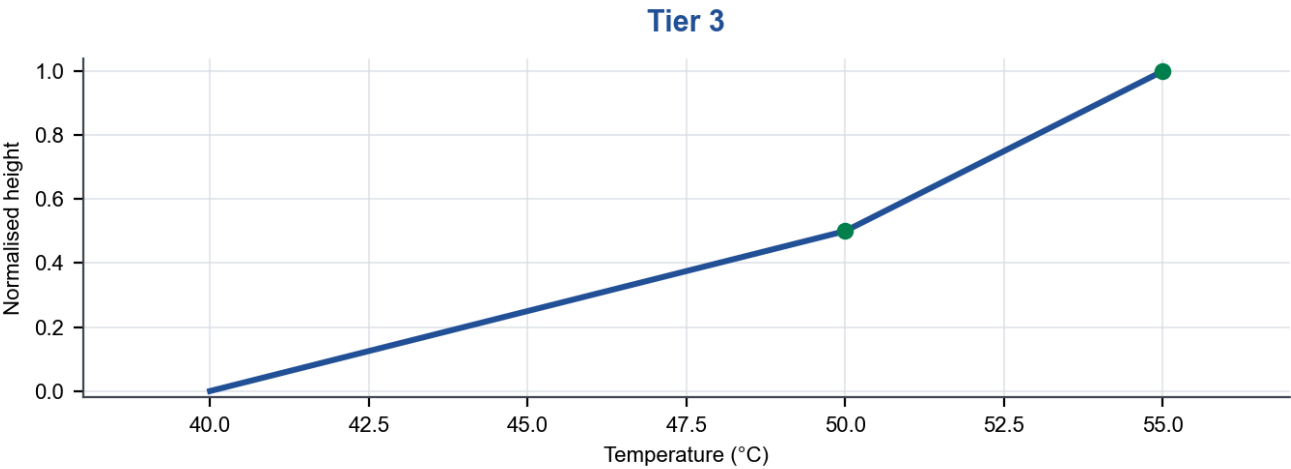
Description	Part No.	Qty	W each	Total (W)
Auxiliary drive	D100	1	100.0	100.0

Tier heat subtotal (W)

300.0

3.1 Temperature Rise Summary — Tier 3

3.1.1 Temperature Rise Curve



3.1.2 Temperature Rise Results

Parameter	Value
Tier heat load (W)	300.0
Effective cooling area Ae (m²)	4.500
Final Temp @ 0.5t (°C)	50.0
Final Temp @ 1.0t (°C)	55.0
Maximum Allowed (°C)	70.0
Compliance @ 0.5t	Compliant
Compliance @ 1.0t	Compliant

Important application of results

Compliance is conditional on providing and maintaining the stated delivered airflow through the tier. A non-compliant result requires additional cooling or a design change. Enclosure temperature compliance does not establish full rated-current capacity; component current capacities and temperature limits also apply.

- Tier 3:** Selected fan: Custom fan arrangement; delivers 480 m³/h of airflow.

3.2 IEC 60890 Calculation Sheet — Tier 3

The following tables summarise the IEC 60890 enclosure thermal calculation including effective cooling surfaces, enclosure constants, and heat dissipation balance.

3.2.1 Effective cooling surfaces

Surface	Dimensions (m)	A0 (m²)	b	Ae (m²)
Front	0.60 × 1.80	1.08	0.90	0.97
			Total Ae	4.50

3.2.2 IEC 60890 calculation parameters

Parameter	Value
Effective area Ae (m²)	4.500
k (enclosure constant)	0.150
c (distribution factor)	1.400
x (exponent factor)	0.804
f (Ae > 1.25 m²)	1.000
Ambient temperature (°C)	40.0

3.2.3 Enclosure heat dissipation and ventilation

Parameter	Value
Is vented (openings specified)	No
Vent inlet opening area (cm²)	N/A
Maximum natural enclosure dissipation P890 (W)	180.0 W
Heat for ventilation / cooling (W)	120.0 W
Required airflow (m³/h)	—
Horizontal partitions	Disabled
Selected fan	Custom fan arrangement
Delivered airflow (m³/h)	480

3.3 Component Current Capacity Assessment - Tier 3

3.3.1 Component derated current capacity

Component	Rated (A)	Internal temp (°C)	Derated output (A)
Auxiliary drive*	100.0	55.0	85.0

* Component(s) temperature deratings were solved directly using manufacturer data. Selected fan: Custom fan arrangement; delivers 480 m³/h of airflow.

Current capacity evaluated at solved top air temperature (55.0°C). Component temperature ratings must also be satisfied.